CO2 ALM1

Seminar

1. Justify why SURF uses integral images for fast feature computation and identify factors affecting the number of detected keypoints. Differentiate SURF and SIFT in terms of descriptor dimensionality and matching efficiency.
2. Distinguish between Sliding Window and Selective Search approaches in object detection and evaluate how Selective Search reduces computational redundancy while maintaining object coverage.
3. Demonstrate SURF orientation assignment and explain how rotation invariance is achieved using Haar wavelet responses.
4. Compare HOG and SIFT descriptors in terms of feature representation, robustness, and computational efficiency.
5. Illustrate HOG descriptor formation including gradient computation, orientation binning, and histogram voting, and explain how block normalization improves robustness.
6. Construct the SIFT scale-space formation process and explain how Difference-of-Gaussian extrema lead to stable keypoints.
7. Describe how Selective Search hierarchically merges regions based on color, texture, and size similarities.
8. Relate deformation cost in a Deformable Part-based Model to part displacement and detection score.
9. Explain how IoU, precision, and recall metrics classify detection outputs as TP, FP, or FN.
10. Assess the impact of Hessian threshold variation on SURF keypoint density and detection speed.
11. Demonstrate SURF feature matching with BFMatcher and ratio test for recognition of rotated and scaled objects.
12. Explain how SURF-based mobile object recognition handles rotated and scaled images for robust identification.

CO2 ALM2

Case Study

1. Calculate the number of sliding windows for a 300×300 image with a 64×64 window and stride 32, and determine the total windows for a 400×400 image using a 128×128 window and stride 64.
2. Compute the Intersection over Union (IoU) for bounding boxes A(20,20,100,100) and B(40,40,120,120), and explain its significance in detection evaluation.
3. Apply SURF feature matching in a mobile application for rotated and scaled object recognition.
4. A robotic vision system must detect multiple objects under scale and rotation variations. Explain how SIFT feature matching supports reliable detection.
5. Apply Selective Search region merging in a cluttered scene for object proposal generation.